

RESHMA THOMAS

AI / ML ENGINEER — AGENTIC AI & AUTOMATION

Kochi, Kerala, India · reshma.thomas.nobel@gmail.com · [Portfolio](#) · [LinkedIn](#) · [GitHub](#)

SUMMARY

AI/ML Engineer building production AI agents — agentic RAG, multi-agent orchestration, and workflow automation. Shipped a labour-law RAG assistant that is live and used daily by a corporate HR team in Doha. Backed by depth in computer vision and document intelligence, plus 8+ years running multilingual editorial operations across 50+ contributors — where I delivered a 40% cut in processing time through workflow automation. Available for contract engagements.

SELECTED PROJECTS

Qatar Labour Law — Agentic RAG Assistant

LIVE IN PRODUCTION

- Built and deployed an agentic RAG system answering Qatar labour law queries with the exact article cited — in daily use by a corporate HR team in Doha.
- Architected retrieval as a tool the model chooses: it searches, evaluates sufficiency, and re-queries for multi-step legal questions rather than performing a single blind lookup.
- Designed three custom tools — semantic topic search, deterministic article-number lookup (semantic search is unreliable for exact references), and side-by-side article comparison for cross-provision reasoning.
- Stack: Claude Tool Use API · ChromaDB · PyMuPDF · Streamlit · Vector Search

Saksham — Four-Agent Assistive Communication System

GOOGLE ADK

- Architected four specialised agents: perception (OpenCV), need detection (Gemini Vision), natural language generation, and personalized TTS output.
- Built confidence-gated dynamic orchestration — ≥ 0.70 proceeds autonomously, < 0.70 escalates to a human, < 0.50 loops back to recapture, since at very low confidence the input is the problem rather than the reasoning.
- Exposed tools via an MCP server for portability; validated against a 5-scenario eval suite achieving 100% task success at 93% average confidence.
- Stack: Google ADK · Gemini Vision · MCP · OpenCV · Multi-Agent Orchestration · Human-in-the-Loop

EXPERIENCE

AI/ML Engineer Intern · iHub Research & Robotics, Kochi

OCT 2024 – PRESENT

- Led development of a medicine blister OCR and Vision-Language Model pipeline enabling robotic pharmacy automation; independently researched open-source frontier models and shaped architectural decisions.
- Developed YOLO-based computer vision pipelines for semi-humanoid robotics — curated datasets, trained models, and benchmarked variants to optimise inference latency for production.

Contributing Editor & Editorial Council Member · Shalom Tidings, Shalom Media USA

JAN 2017 – PRESENT

- Designed and deployed automated content workflows processing 1,000+ documents annually across 12 languages — delivering a 40% reduction in processing time.
- Coordinate 50+ global contributors across 12 languages, shipping a print publication on a fixed deadline every cycle.
- Spearheaded editorial standards contributing to multiple international press awards: Gold Award, Australian Religious Press Association (2020); Best Magazine of the Year, Catholic Press Awards USA (2021, 2022, 2023).

TECHNICAL SKILLS

Agentic AI & Automation Claude Tool Use API · Google ADK · MCP · Agentic RAG · Multi-Agent Orchestration · ChromaDB · Vector Search · Human-in-the-Loop Design

LLMs & GenAI Claude API · Gemini 1.5 Pro · Gemini Vision · OpenAI API · Prompt Engineering · RAG · Fine-tuning · Qwen 2.5 VLM

ML / CV / OCR Python · PyTorch · TensorFlow · Keras · Scikit-learn · OpenCV · YOLO · MobileNet · PaddleOCR · EasyOCR · Roboflow

Backend & Deployment FastAPI · Flask · Django · Streamlit · Docker · REST APIs · Git · SQL

EDUCATION & CERTIFICATIONS

Post Graduate Diploma in AI & Machine Learning · iHub School of Learning

2025 – 2026

B.Tech in Biotechnology — First Class with Honours · Calicut University, Kerala

COMPLETED

Agentic AI — DeepLearning.AI · Machine Learning Specialization — Stanford University / DeepLearning.AI (Coursera)